Trainings

General Information

This section provides requirements and recommendations for selecting engine coolant, based on the appropriate Cummins® Engineering Standard (CES), for use in engines manufactured by Cummins Inc. This section also provides general information on the function, composition, maintenance, and testing of engine coolant. Reference corresponding engine service literature for additional information.

Not following information located in a requirements section can affect the ability to claim warranty. Information found in the recommendation sections is intended to provide optimal engine protection.

Proper coolant selection, and maintaining the correct coolant testing and maintenance interval are vital factors in preserving the integrity of an engine.

Table 1, Definition of Terms

Term	Definition
Supplemental Coolant Additive (SCA)	Additive chemicals added to engine coolant at regular intervals. Benefits include cylinder liner cavitation protection, pH buffering capacity, and scale prevention. Cummins Filtration™ sells SCA products DCA-2 and DCA-4.
Coolant	As used in this bulletin, coolant refers to liquid mixture in the engine, or vehicle cooling system, that functions to maintain an engine temperature in the designed range. Coolant is made up of water, glycol, and additives; also referred to as "Prediluted" or "Premix".
Antifreeze	Glycol and additive portion of coolant. Controls corrosion and freezing/boiling point of coolant.
Heavy Duty Coolant	Coolant contains correct amount of type of additives to be used in a heavy duty engine. Heavy duty antifreeze/coolant meets American Society for Testing and Materials (ASTM) D6210, at minimum.

Table 1, Definition of Terms	
Term	**Definition**
Partially Formulated	Antifreeze or coolant that requires a "precharge" of SCA to protect against liner pitting and hot surface scaling. Partially formulated antifreeze/coolant does **not** meet ASTM D6210. Partially formulated coolants may also be referred to as "universal coolants."
Treated Water Coolant	Water containing all additives necessary for use as a coolant in heavy duty engines. "treated water" is defined, in this document, as coolant containing greater than 60 percent water by volume.
SCA Unit	SCA concentration is calculated based on the concentration of nitrite and/or molybdate present in a coolant sample. Other coolant additives are **not** included in SCA calculation. SCA units are **not** applicable to organic acid technology (OAT)/Extended Life Coolants.

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